

Co-Midis Tablets 80/5mg

1. NAME OF THE MEDICINAL PRODUCT

Co-Midis Tablets 80/5mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 80 mg telmisartan and 5 mg amlodipine (as amlodipine besilate).

3. PHARMACEUTICAL FORM

Each tablet occurs as a white tablet of oval, biconvex shape; engraved with "747" on one side and " SCP " on the other side.

The tablets are for oral administration.

4. CLINICAL PARTICULARS

4.1 Indications

Treatment of essential hypertension in adults.

Replacement Therapy

Patients receiving telmisartan and amlodipine from separate tablets may instead receive Co-Midis containing the same component doses.

Add-on therapy

Co-Midis is indicated in patients whose blood pressure is not adequately controlled on telmisartan or amlodipine monotherapy.

Initial therapy

Co-Midis may also be used as initial therapy in patients who are likely to need multiple drugs to achieve their blood pressure goals.

Base on the choice of Co-Midis as initial therapy for hypertension on an assessment of potential benefits and risks including whether the patient is likely to tolerate the starting dose of Co-Midis tablets.

Patients with moderate or severe hypertension are at relatively high risk for cardiovascular events (such as stroke, heart attacks and heart failure), kidney failure and vision problems, so prompt treatment is clinically relevant. Consider the patient's baseline blood pressure, the target goals and the incremental likelihood of achieving goals with a combination compared to monotherapy when deciding whether to use Co-Midis tablets as initial therapy. Individual blood pressure goals may vary based upon the patient's risk.

4.2 Dosage and Administration

DOSAGE

Adults

Co-Midis should be taken once daily. The maximum recommendation dose is Telmisartan & Amlodipine 80mg/10mg one tablet per day.

Replacement Therapy

Patients receiving telmisartan and amlodipine from separate tablets can instead receive Co-Midis containing the same component doses in one tablet once daily, e.g. to enhance convenience or compliance.

Add on therapy

Co-Midis may be administered in patients whose blood pressure is not adequately controlled with amlodipine or telmisartan alone.

Patients treated with 10 mg amlodipine who experience any dose limiting adverse reactions such as oedema, may be switched to Telmisartan & Amlodipine 40/5mg once daily, reducing the dose of amlodipine without reducing the overall expected antihypertensive response.

Individual dose titration with the components (ie. amlodipine and telmisartan) is recommended before changing to the fixed dose combination. When clinically appropriate, direct change from monotherapy to the fixed dose combination may be considered.

Initial therapy

A patient may be initiated on Co-Midis if it is unlikely that control of blood pressure would be achieved with a single agent. The usual starting dose of Telmisartan & Amlodipine is 40/5 mg once daily. Patients requiring larger blood pressure reductions may be started on Co-Midis 80/5 mg once daily.

If additional blood pressure lowering is needed after at least 2 weeks of therapy, the dose may be titrated up to a maximum of 80/10 mg once daily.

Initial therapy with Co-Midis is not recommended in patients ≥ 75 years old or with hepatic impairment.

Correct imbalances on intravascular volume- or salt-depletion, before initiating therapy with Co-Midis tablets.

Co-Midis can be administered with other antihypertensive drugs.

Special populations

Geriatric patients

No dose adjustment is necessary for geriatric patients. Little information is available in the very elderly patients.

Normal amlodipine dosage regimens are recommended in the elderly, but increase of dosage should take place with care (see sections Special warnings and precautions, and Pharmacokinetics).

Paediatric patients

Co-Midis is not recommended for use in patients aged below 18 years due to a lack of data on safety and efficacy.

Renal impairment

No posology adjustment is required for patients with renal impairment, including those on haemodialysis. Limited experience is available with severe renal impairment or haemodialysis. Caution is advised when using Co-Midis in such patients as amlodipine and telmisartan are not dialysable. Telmisartan is not removed from blood by hemofiltration and is not dialyzable. Amlodipine is not dialyzable.

Hepatic impairment

In patients with mild to moderate hepatic impairment Co-Midis should be administered with caution. For telmisartan the posology should not exceed 40 mg once daily (see section Contraindications).

Co-Midis 80/5 mg is not able to deliver all approved dose regimens of Telmisartan & Amlodipine. Therefore, other approved dosage forms and strengths of Telmisartan & Amlodipine should be used in such cases.

Method of administration

Co-Midis tablets are for once-daily oral administration and should be swallowed whole with liquid.

Co-Midis can be taken with or without food.

Handling Instructions

Due to the hygroscopic property of the tablets they should be taken out of the sealed blister shortly before administration

4.3 Contraindications

- ☐ Hypersensitivity to the active substances, or to any of the excipients.
- ☐ Hypersensitivity to dihydropyridine derivatives
- ☐ Second and third trimesters of pregnancy
- ☐ Lactation
- ☐ Biliary obstructive disorders
- ☐ Severe hepatic impairment

- ☐ Severe hypotension
- ☐ Shock (including cardiogenic shock)
- ☐ Obstruction of the outflow tract of the left ventricle (e.g. high grade aortic stenosis)
- ☐ Haemodynamically unstable heart failure after acute myocardial infarction
- ☐ The concomitant use of Co-Midis with aliskiren is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73 m²)

In case of rare hereditary conditions that may be incompatible with an excipient of the product (see section "Special warnings and precautions") the use of the product is contraindicated.

4.4 Special Warnings and Precautions

Pregnancy

Angiotensin II receptor antagonists should not be initiated during pregnancy.

Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy.

When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and if appropriate, alternative therapy should be started.

Hepatic impairment

Telmisartan is mostly eliminated in the bile. Patients with biliary obstructive disorders or hepatic insufficiency can be expected to have reduced clearance.

The half-life of amlodipine is prolonged and AUC values are higher in patients with impaired liver function; dosage recommendations have not been established. Amlodipine should therefore be initiated at the lower end of the dosing range and caution should be used, both on initial treatment and when increasing the dose.

Co-Midis should therefore be used with caution in these patients.

Renovascular hypertension

There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system.

Renal impairment and kidney transplant

When Co-Midis is used in patients with impaired renal function, a periodic monitoring of potassium and creatinine serum levels is recommended. There is no experience regarding the administration of Co-Midis in patients with a recent kidney transplant.

Telmisartan and Amlodipine are not dialyzable.

Intravascular hypovolaemia

Symptomatic hypotension, especially after the first dose, may occur in patients who are volume and/or sodium depleted by e.g. vigorous diuretic therapy, dietary salt restriction, diarrhoea or vomiting. Such conditions should be corrected before the administration of Co-Midis.

Dual blockade of the renin-angiotensin-aldosterone system (RAAS)

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren increases the risk of hypotension, hyperkalaemia and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended (see Interactions).

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

Other conditions with stimulation of the renin-angiotensin-aldosterone system

In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with medicinal products that affect this system has been associated with acute hypotension, hyperazotaemia, oliguria, or rarely acute renal failure.

Primary aldosteronism

Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of telmisartan is not recommended.

Aortic and mitral valve stenosis, obstructive hypertrophic cardiomyopathy

As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Unstable angina pectoris, acute myocardial infarction

There are no data to support the use of Co-Midis in unstable angina pectoris and during or within one month of a myocardial infarction.

Patients with cardiac failure

In an amlodipine long-term, placebo controlled study in patients with severe heart failure (NYHA class III and IV) the reported incidence of pulmonary oedema was higher in the amlodipine treated group than in the placebo group. Therefore, patients with heart failure should be treated with caution.

Calcium channel blockers, including amlodipine, should be used with caution in patients with congestive heart failure, as they may increase the risk of future cardiovascular events and mortality.

Hyperkalaemia

During treatment with medicinal products that affect the renin-angiotensin-aldosterone system hyperkalaemia may occur, especially in the presence of renal impairment and/or heart failure. Monitoring of serum potassium in patients at risk is recommended.

Based on experience with the use of medicinal products that affect the renin-angiotensin system, concomitant use with potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium or other medicinal products that may increase the potassium level (heparin, etc.) may lead to an increase in serum potassium and should therefore be co-administered cautiously with telmisartan.

Sorbitol

Co-Midis tablets 80/5 mg contains 337.28 mg in each tablet.

Sorbitol is a source of fructose. Co-Midis tablets 80/5 mg is not recommended for use in patients with hereditary fructose intolerance.

Diabetes mellitus

In diabetic patients with an additional cardiovascular risk, i.e. patients with diabetes mellitus and coexistent coronary artery disease (CAD), the risk of fatal myocardial infarction and unexpected cardiovascular death may be increased when treated with blood pressure lowering agents such as ARBs or ACE-inhibitors. In patients with diabetes mellitus CAD may be asymptomatic and therefore undiagnosed. Patients with diabetes mellitus should undergo appropriate diagnostic evaluation, e.g. exercise stress testing, to detect and to treat CAD accordingly before initiating treatment with Co-Midis.

Elderly patients

The increase of the amlodipine dosage should take place with care in the elderly patients (see sections Dosage and Administration and Pharmacokinetics)

Other

Co-Midis was effective when treating black patients (usually a low-renin population).

As with any antihypertensive agent, excessive reduction of blood pressure in patients with ischaemic cardiopathy or ischaemic cardiovascular disease could result in a myocardial infarction or stroke.

4.5 Fertility, Pregnancy and Lactation

The effects of Co-Midis during pregnancy and lactation are not known. Effects related to the mono components are described below.

Pregnancy

Telmisartan:

The use of angiotensin II receptor antagonists is not recommended during the first trimester of pregnancy and should not be initiated during pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started. The use of angiotensin II receptor antagonists is contraindicated during the second and third trimester of pregnancy. Preclinical studies with telmisartan do not indicate teratogenic effect, but have shown fetotoxicity. Angiotensin II receptor antagonists exposure during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy.

Should exposure to angiotensin II receptor antagonists have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.

Infants whose mothers have taken angiotensin II receptor antagonists should be closely observed for hypotension.

Amlodipine:

The safety of amlodipine in human pregnancy has not been established. In animal studies, reproductive toxicity was observed at high doses.

Lactation:

Co-Midis is contraindicated during lactation since it is not known whether telmisartan is excreted in human milk. Animal studies have shown excretion of telmisartan in breast milk. Amlodipine is excreted in human milk. The proportion of the maternal dose received by the infant has been estimated with an interquartile range of 3-7%, with a maximum of 15%. The effect of amlodipine on infants is unknown. Because of the potential adverse reactions in nursing infants, a decision should be made whether to discontinue nursing or to discontinue therapy, taking into account the importance of this therapy for the mother (see section contraindication).

Fertility:

No data from controlled clinical studies with the fixed dose combination or with the individual components are available.

Separate reproductive toxicity studies with the combination of telmisartan and amlodipine have not been conducted. In preclinical studies, no effects of telmisartan on male and female fertility were observed. In some patients treated by calcium channel blockers, reversible biochemical changes in the head of spermatozoa have been reported. Clinical data are insufficient regarding the potential effect of amlodipine on fertility. In one rat study, adverse effects were found on male fertility.

4.6 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However, patients should be advised that they may experience undesirable effects such as syncope, somnolence, dizziness, or vertigo during treatment. Therefore, caution should be recommended when driving a car or operating machinery.

If patients experience these adverse experiences, they should avoid potentially hazardous tasks such as driving or operating machinery.

4.7 Side effects

Tabulated summary of adverse reactions

The following adverse reactions derived from the use of the telmisartan/amlodipine combination or the use of the monocomponents (telmisartan or amlodipine) in clinical trials or from post-marketing experience are shown below classified by MedDRA System organ class and MedDRA Preferred terms.

Infections and infestations

Cystitis

Psychiatric disorders

Depression, anxiety, insomnia

Nervous system disorders

Dizziness, somnolence, migraine, headache, paraesthesia, syncope, neuropathy peripheral, hypoaesthesia, dysgeusia, tremor

Ear and labyrinth disorders

Vertigo

Cardiac disorders

Bradycardia, palpitations

Vascular disorders

Hypotension, orthostatic hypotension, flushing

Respiratory, thoracic and mediastinal disorders

Cough

Gastrointestinal disorders

Abdominal pain, diarrhoea, nausea, vomiting, gingival hypertrophy, dyspepsia, dry mouth

Skin and subcutaneous tissue disorders

Pruritus, eczema, erythema, rash

Musculoskeletal and connective tissue disorders

Arthralgia, muscle spasms (cramps in legs), myalgia, back pain, pain in extremity (leg pain)

Renal and urinary disorders

Nocturia

Reproductive system and breast disorders

Erectile dysfunction

General disorders

Oedema Peripheral, asthenia (weakness), chest pain, fatigue, oedema, malaise

Investigations

Hepatic enzyme increased, blood uric acid increased

Telmisartan:

Infections and infestations

Urinary tract infection, upper respiratory tract infection, sepsis including fatal outcome

Blood and lymphatic system disorders

Anaemia, eosinophilia, thrombocytopenia

Immune system disorders

Anaphylactic reaction, hypersensitivity

Metabolism and nutrition disorders

Hyperkalaemia, hypoglycaemia (in diabetic patients)

Eye disorders

Visual impairment

Cardiac disorders

Tachycardia

Respiratory, thoracic and mediastinal disorders

Dyspnoea

Gastrointestinal disorders

Flatulence, abdominal discomfort

Hepatobiliary disorders

Hepatic function abnormal/ liver disorder*

*Most cases of hepatic function abnormal / liver disorder from post-marketing experience with telmisartan occurred in patients in Japan, who are more likely to experience these adverse reactions

Skin and subcutaneous tissue disorders

Angioedema (with fatal outcome), hyperhidrosis, urticaria, drug eruption, toxic skin eruption

Musculoskeletal and connective tissue disorders

Tendon pain (tendonitis like symptoms)

Renal and urinary disorders

Renal impairment including renal impairment (including acute kidney injury)(see also Special warnings and precautions)

General disorders

Influenza like illness

Investigations

Blood creatinine increased, haemoglobin decreased, blood creatine phosphokinase (CPK) increased

Amlodipine:

Blood and lymphatic system disorders

Leukopenia, thrombocytopenia

Immune system disorders

Hypersensitivity

Metabolism and nutrition disorders

Hyperglycaemia

Psychiatric disorders

Mood altered, confusional state

Nervous system disorders

Extrapyramidal disorder, Hypertonia

Eye disorders

Visual impairment, Diplopia

Ear and labyrinth disorders

Tinnitus

Cardiac disorders

Myocardial infarction, arrhythmia, ventricular tachycardia, atrial fibrillation

Vascular disorders

Vasculitis

Respiratory, thoracic and mediastinal disorders:

Dyspnoea, rhinitis

Gastrointestinal disorders

Change of bowel habit, pancreatitis, gastritis, Constipation

Hepatobiliary disorders

Hepatitis, jaundice, hepatic enzyme increase (mostly consistent with cholestasis)

Skin and subcutaneous tissue disorders

Hyperhidrosis, angioedema, urticaria, alopecia, purpura, skin discolouration, erythema multiforme, dermatitis exfoliative, Stevens-Johnson syndrome, photosensitivity reaction, Toxic epidermal necrolysis

Renal and urinary disorders

Micturition disorder, pollakiuria

Reproductive system and breast disorders

Gynaecomastia

General disorders

Pain, weight increased, weight decreased

Musculoskeletal and connective tissue disorders

Joint Swelling

4.8 Drug Interactions

No interactions between the two components of this fixed dose combinations have been observed in clinical studies.

Agents with blood pressure lowering potential

Based on their pharmacological properties it can be expected that the following medicinal products may potentiate the hypotensive effects of all antihypertensives including Co-Midis, e.g. baclofen, amifostine. Furthermore, orthostatic hypotension may be aggravated by alcohol, barbiturates, narcotics, or antidepressants.

Corticosteroids (systemic route)

Reduction of the antihypertensive effect.

Interactions linked to telmisartan

Telmisartan may increase the hypotensive effect of other antihypertensive agents.

Co-administration of telmisartan did not result in a clinically significant interaction with digoxin, warfarin, hydrochlorothiazide, glibenclamide, ibuprofen, paracetamol, simvastatin and amlodipine. For digoxin a 20% increase in median plasma digoxin trough concentration has been observed (39% in a single case), monitoring of plasma digoxin levels should be considered.

In one study the co-administration of telmisartan and ramipril led to an increase of up to 2.5 fold in the AUC₀₋₂₄ and C_{max} of ramipril and ramiprilat. The clinical relevance of this observation is not known. Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors. Cases have also been reported with angiotensin II receptor antagonists including telmisartan. Therefore, serum lithium level monitoring is advisable during concomitant use.

Treatment with NSAIDs (i.e. ASA at anti-inflammatory dosage regimens, COX-2 inhibitors and non-selective NSAIDs) is associated with the potential for acute renal insufficiency in patients who are dehydrated. Compounds acting on the Renin-Angiotensin-System like telmisartan may have synergistic effects. Patients receiving NSAIDs and telmisartan should be adequately hydrated and be monitored for renal function at the beginning of combined treatment.

A reduced effect of antihypertensive drugs like telmisartan by inhibition of vasodilating prostaglandins has been reported during combined treatment with NSAIDs.

Clinical trial data has shown that dual blockade of the renin-angiotensin-aldosterone-system (RAAS) through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is associated with a higher frequency of adverse events such as hypotension, hyperkalaemia and decreased renal function (including acute renal failure) compared to the use of a single RAAS-acting agent (see Contraindications and Warnings/Precautions).

Aliskiren: Do not co-administer aliskiren with Co-Midis in patients with diabetes. Avoid use of aliskiren with Co-Midis in patient with renal impairment (GFR < 60 ml/min).

Interactions linked to amlodipine

Grapefruit and grapefruit juice

Administration of Co-Midis with grapefruit or grapefruit juice is not recommended since bioavailability may be increased in certain patients resulting in increased blood pressure lowering effects.

CYP3A4 inhibitors

Concomitant use of amlodipine with strong or moderate CYP3A4 inhibitors (protease inhibitors, azole antifungals, macrolides like erythromycin or clarithromycin, verapamil or diltiazem) may give rise to significant increase in amlodipine exposure resulting in an increased risk of hypotension. The clinical translation of these PK variations may be more pronounced in the elderly. Clinical monitoring and dose adjustment may thus be required.

CYP3A4 inducers

Upon co-administration of known inducers of the CYP3A4, the plasma concentration of amlodipine may vary. Therefore, blood pressure should be monitored and dose regulation considered both during and after concomitant medication particularly with strong CYP3A4 inducers (e.g. rifampicin, hypericum perforatum).

Dantrolene (infusion)

In animals, lethal ventricular fibrillation and cardiovascular collapse are observed in association with hyperkalemia after administration of verapamil and intravenous dantrolene. Due to risk of hyperkalemia, it is recommended that the coadministration of calcium channel blockers such as amlodipine be avoided in patients susceptible to malignant hyperthermia and in the management of malignant hyperthermia.

Simvastatin

Co-administration of multiple doses of 10mg of amlodipine with simvastatin 80 mg resulted in an increase in exposure to simvastatin up to 77% compared to simvastatin alone. Therefore, limit the dose of simvastatin in patients on amlodipine to 20 mg daily.

Tacrolimus

There is a risk of increased tacrolimus blood levels when co-administered with amlodipine but the pharmacokinetic mechanism of this interaction is not fully understood. In order to avoid toxicity of tacrolimus, administration of amlodipine in a patient treated with tacrolimus requires monitoring of tacrolimus blood levels and dose adjustment of tacrolimus when appropriate.

Cyclosporine

No drug interaction studies have been conducted with cyclosporine and amlodipine in healthy volunteers or other populations with the exception of renal transplant patients, where variable trough concentration increases (average 0% - 40%) of cyclosporine were observed. Consideration should be given for monitoring cyclosporine levels in renal transplant patients on amlodipine, and cyclosporine dose reductions should be made as necessary.

Mechanistic Target of Rapamycin (mTOR) Inhibitors

mTOR inhibitors such as sirolimus, temsirolimus, and everolimus are CYP3A substrates. Amlodipine is a weak CYP3A inhibitor. With concomitant use of mTOR inhibitors, amlodipine may increase exposure of mTOR inhibitors

Additional information

In clinical interaction studies, amlodipine did not affect the pharmacokinetics of atorvastatin, digoxin or warfarin.

4.9 Overdose

Symptoms

There is no experience of overdose with Co-Midis. Signs and symptoms of overdose are expected to be in line with exaggerated pharmacological effects.

The most prominent manifestations of telmisartan overdosage were hypotension, tachycardia; bradycardia also occurred. Overdose with amlodipine may result in excessive peripheral vasodilatation and possibly reflex tachycardia. Marked and probably prolonged systemic hypotension up to and including shock with fatal outcome may occur.

Therapy

Supportive treatment should be instituted. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade.

Gastric lavage may be worthwhile in some cases. In healthy volunteers the use of charcoal up to 2 hours after administration of amlodipine 10 mg has been shown to reduce the absorption rate of amlodipine.

Telmisartan and amlodipine are not removed by haemodialysis .

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamics properties

Pharmacotherapeutic group: angiotensin II antagonists, plain (telmisartan), combinations with dihydropyridine derivatives (amlodipine), ATC Code: C09DB04.

Mode of action

Co-Midis combines two antihypertensive compounds with complementary mechanisms to control blood pressure in patients with essential hypertension: an angiotensin II receptor antagonist, telmisartan, and a dihydropyridinic calcium channel blocker, amlodipine.

The combination of these substances has an additive antihypertensive effect, reducing blood pressure to a greater degree than either component alone.

Co-Midis once daily produces effective and consistent reductions in blood pressure across the 24-hour therapeutic dose range.

Telmisartan:

Telmisartan is an orally effective and specific angiotensin II receptor (type AT1) antagonist. Telmisartan displaces angiotensin II with very high affinity from its binding site at the AT1 receptor subtype, which is responsible for the known actions of angiotensin II. Telmisartan does not exhibit any partial agonist activity at the AT1 receptor. Telmisartan selectively binds the AT1 receptor. The binding is long lasting.

Telmisartan does not show affinity for other receptors, including AT2 and other less characterised AT receptors. The functional role of these receptors is not known, nor is the effect of their possible overstimulation by angiotensin II, whose levels are increased by telmisartan. Plasma aldosterone levels are decreased by telmisartan. Telmisartan does not inhibit human plasma renin or block ion channels.

Telmisartan does not inhibit angiotensin converting enzyme (kininase II), the enzyme which also degrades bradykinin. Therefore it is not expected to potentiate bradykinin-mediated adverse effects.

In man, an 80 mg dose of telmisartan almost completely inhibits the angiotensin II evoked blood pressure increase. The inhibitory effect is maintained over 24 hours and still measurable up to 48 hours.

Amlodipine:

Amlodipine is a calcium ion influx inhibitor of the dihydropyridine group (slow channel blocker or calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into cardiac and vascular smooth muscle.

The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on vascular smooth muscle, leading to reductions in peripheral vascular resistance and in blood pressure. Experimental data indicate that amlodipine binds to both dihydropyridine and non-dihydropyridine binding sites. Amlodipine is relatively vessel-selective, with a greater effect on vascular smooth muscle cells than on cardiac muscle cells.

Pharmacodynamics

Telmisartan:

After the first dose of telmisartan, the antihypertensive activity gradually becomes evident within 3 hours. The maximum reduction in blood pressure is generally attained 4 weeks after the start of treatment and is sustained during long-term therapy.

The antihypertensive effect persists constantly over 24 hours after dosing and includes the last 4 hours before the next dose as shown by ambulatory blood pressure measurements. This is confirmed by trough to peak ratios consistently above

80 % seen after doses of 40 and 80 mg of telmisartan in placebo controlled clinical studies.

There is an apparent trend to a dose relationship to a time to recovery of baseline SBP. In this respect data concerning DBP are inconsistent.

In patients with hypertension telmisartan reduces both systolic and diastolic blood pressure without affecting pulse rate. The antihypertensive efficacy of telmisartan is comparable to that of agents representative of other classes of antihypertensive drugs (demonstrated in clinical trials comparing telmisartan to amlodipine, atenolol, enalapril, hydrochlorothiazide, losartan, lisinopril, ramipril and valsartan).

Upon abrupt cessation of treatment with telmisartan, blood pressure gradually returns to pre-treatment values over a period of several days without evidence of rebound hypertension.

Telmisartan treatment has been shown in clinical trials to be associated with statistically significant reductions in Left Ventricular Mass and Left Ventricular Mass Index in patients with hypertension and Left Ventricular Hypertrophy. Telmisartan treatment has been shown in clinical trials (including comparators like losartan, ramipril and valsartan) to be associated with statistically significant reductions in proteinuria (including microalbuminuria and macroalbuminuria) in patients with hypertension and diabetic nephropathy. The incidence of dry cough was significantly lower in patients treated with telmisartan than in those given angiotensin converting enzyme inhibitors in clinical trials directly comparing the two antihypertensive treatments.

Amlodipine:

In patients with hypertension, once daily dosing provides clinically significant reductions of blood pressure in both the supine and standing positions throughout the 24 hour interval. Due to the slow onset of action, acute hypotension is not a feature of amlodipine administration.

In hypertensive patients with normal renal function, therapeutic doses of amlodipine resulted in a decrease in renal vascular resistance and an increase in glomerular filtration rate and effective renal plasma flow, without change in filtration fraction or proteinuria

5.2 Pharmacokinetics properties

Pharmacokinetics of the fixed dose combination

The rate and extent of absorption of Co-Midis are equivalent to the bioavailability of telmisartan and amlodipine when administered as individual tablets.

Pharmacokinetic of the single components:

Absorption

Absorption of telmisartan is rapid although the amount absorbed varies. The mean absolute bioavailability for telmisartan is about 50%.

When telmisartan is taken with food, the reduction in the area under the plasma concentration-time curve (AUC) of telmisartan varies from approximately 6% (40 mg dose) to approximately 19% (160 mg dose). By 3 hours after administration plasma concentrations are similar whether telmisartan is taken fasting or with food.

The small reduction in AUC is not expected to cause a reduction in the therapeutic efficacy.

After oral administration of therapeutic doses of amlodipine alone, peak plasma concentrations of amlodipine are reached in 6–12 hours. Absolute bioavailability has been calculated as between 64% and 80%. Amlodipine bioavailability is unaffected by food ingestion.

Distribution:

Telmisartan is largely bound to plasma protein (> 99.5 %), mainly albumin and alpha-1 acid glycoprotein. The mean steady state apparent volume of distribution (V_{ss}) is approximately 500 L.

The volume of distribution of amlodipine is approximately 21 L/kg. In vitro studies with amlodipine have shown that approximately 97.5% of circulating drug is bound to plasma proteins in hypertensive patients.

Metabolism:

Telmisartan is metabolised by conjugation to the glucuronide of the parent compound. No pharmacological activity has been shown for the conjugate.

Amlodipine is extensively (approximately 90%) metabolised by the liver to inactive metabolites.

Elimination:

Telmisartan is characterised by biexponential decay pharmacokinetics with a terminal elimination half-life of > 20 hours. The maximum plasma concentration (C_{max}) and, to a smaller extent, area under the plasma concentration-time curve (AUC) increase disproportionately with dose. There is no evidence of clinically relevant accumulation of telmisartan. After oral (and intravenous) administration telmisartan is nearly exclusively excreted with the faeces, exclusively as unchanged compound. Cumulative urinary excretion is < 2% of dose. Total plasma clearance (CL_{tot}) is high (approximately 900 ml/min compared with hepatic blood flow (about 1500 mL/min)).

Amlodipine elimination from plasma is biphasic, with a terminal elimination half-life of approximately 30 to 50 hours. Steady-state plasma levels are reached after continuous administration for 7–8 days. Ten per cent of original amlodipine and 60% of amlodipine metabolites are excreted in urine.

Paediatric population (age below 18 years)

No pharmacokinetic data for Co-Midis are available in the paediatric population.

Gender effects:

Gender differences in plasma concentrations of telmisartan were observed, C_{max} and AUC being approximately 3- and 2- fold higher, respectively, in females compared to males without relevant influence on efficacy.

Elderly patients

The pharmacokinetics of telmisartan do not differ between younger and elderly patients.

Time to peak plasma amlodipine concentrations is similar in young and elderly patients. In elderly patients, amlodipine clearance tends to decline, causing increases in the area under the curve (AUC) and elimination half-life.

Renal impairment

Lower plasma concentrations of telmisartan were observed in patients with renal insufficiency undergoing dialysis. Telmisartan is highly bound to plasma protein in renal-insufficient subjects and cannot be removed by dialysis. The elimination half-life is not changed in patients with renal impairment. The pharmacokinetics of amlodipine are not significantly influenced by renal impairment.

Patients with hepatic impairment

Pharmacokinetic studies in patients with hepatic impairment showed an increase in absolute bioavailability of telmisartan up to nearly 100%. The elimination half-life is not changed in patients with hepatic impairment. Patients with hepatic insufficiency have decreased clearance of amlodipine with resulting increase of approximately 40– 60% in AUC .

6. PHARMACEUTICAL PARTICULARS

6.1 Nature and contents of container

3 x 10 tablets in a box.

i.e. There are 10 tablets per blister. There are 3 blister packaging foil (lidding foil)-blister packaging material (Alu-Alu) in one box.

6.4 Storage conditions

Do not store above 30 °C.

Store in the original package in order to protect from light and moisture.

7. NAME AND ADDRESS OF MANUFACTURER

Standard Chem. & Pharm. Co., Ltd.

No. 6-20, Tuku, Tuku Village, Sinying

District, Tainan City 73055, Taiwan

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